

Ankle: Anatomy & How It Works

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What is the ankle joint?

The ankle is the joint that connects your foot to your lower leg. Healthcare providers sometimes refer to it as the tibiotalar joint or the talocrural joint.

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Like all joints, your ankles are part of your skeletal system. Your ankles also contain cartilage, muscles, ligaments and nerves.

Visit a healthcare provider if you're experiencing ankle pain. They'll diagnose what's causing your pain and other symptoms and will suggest treatments that'll help you get back to your usual routine and activities.

Function

What does the ankle joint do?

Your ankles bend and flex anytime you're moving to keep you stable and maintain your balance.

Your ankles move in two directions:

Plantar flexion: Down, away from your body.

Dorsiflexion: Up, toward your body.

Anatomy

Where is the ankle joint located?

The ankle is at the lower end of your leg. It's where your lower leg meets your foot.

Which type of joint is the ankle?

There are lots of joints and many ways healthcare providers group them together. Joints are usually classified based on:

Their composition: What they're made of.

Their function: How they move.

The ankle is a synovial joint. Synovial joints have the most freedom to move. They're made of a cavity in one bone that another bone fits into. Slippery hyaline cartilage covers the ends of bones that make up a synovial joint. A synovial membrane — a fluid-filled sac that lubricates and protects the joint — lines the space between the bones. This extra cushioning helps synovial joints move with as little friction as possible.

Functionally, the ankle is a hinge joint. Think about the hinges that hold a door in place. They have a few parts that don't move, but other pieces move a specific distance to open and close. Your ankle hinges in either direction it can move.

What is the structure of the ankle joint?

Your ankles are made of:

Bones.

Cartilage.

Ligaments.

Muscles.

Nerves.

Blood vessels.

Bones in the ankle

Three bones make up your ankle joint:

Tibia (shin bone).

Fibula (calf bone).

Talus.

Cartilage in the ankle

Cartilage is a strong, flexible tissue that protects your joints. It acts as a shock absorber throughout your body. Your ankle is lined with hyaline cartilage. Hyaline cartilage is the most common type of cartilage in your body. Some healthcare providers call it articular cartilage. It lines your joints and caps the ends of your bones.

Hyaline cartilage is slippery and smooth, which helps your bones move smoothly past each other in your joints. The surfaces of your tibia, fibula and talus that touch each other have a hyaline cartilage lining.

Ligaments in the ankle

Ankle ligaments are like cords that connect bones in your foot to your lower leg bones.

There are three main sets of ligaments in your ankle:

Medial ligaments (deltoid ligaments): These four ligaments start at the medial malleolus (the bottom end of your tibia). They fan out to connect to the talus, calcaneus (heel bone) and navicular bones in your foot.

Lateral ligaments: These three ligaments start at the lateral malleolus — the end of the fibula that forms the bump on the outside of your ankle. They connect to the talus and calcaneus.

Syndesmotc ligaments: These four ligaments connect the tibia and fibula.

Muscles in the ankle

Muscles are soft tissue made of stretchy fibers. They tense up (flex) to pull and move parts of your body. Muscles attached to your legs and feet control your ankle movements.

Muscles that control plantarflexion let you flex your foot down, away from your body. They include the:

Gastrocnemius.

Soleus.

Plantaris.

Tibialis posterior.

Flexor digitorum longus.

Flexor hallucis longus.

Peroneus brevis.

Peroneus longus.

Muscles that control dorsiflexion let you move your foot up toward your body. They include the:

Tibialis anterior.

Extensor digitorum longus.

Extensor hallucis longus.

Peroneus tertius.

Nerves in the ankle

Nerves are like cables that carry electrical impulses between your brain and the rest of your body. These impulses help you feel sensations and move your muscles. Nerves in your ankle include the:

Tibial nerve.

Superficial peroneal nerve.

Deep peroneal nerve.

Blood vessels in the ankle

Blood vessels are channels that carry blood throughout your body. They form a closed loop, like a circuit, that begins and ends at your heart. Three arteries carry blood to and from your ankle, including the:

Anterior tibial artery.

Posterior tibial artery.

Peroneal artery.

Conditions and Disorders

What are common issues or conditions that affect ankles?

Your ankles can be affected by anything that damages your bones or connective tissue, including:

Foot and ankle arthritis.

Gout.

Bursitis.

Tendinitis.

Flat feet.

Sports injuries are some of the most common ankle injuries:

Ankle sprains.

Achilles tendon injuries and other tendon injuries.

Ankle fractures.

What are common symptoms that affect ankle joints?

The condition or issue that's affecting your ankle will determine which symptoms you experience. Many ankle issues cause symptoms like:

Pain (especially when moving).

Inflammation (swelling).

Stiffness.

Instability.

Discoloration or redness around your ankle.

A feeling of warmth or heat.

What are common tests done to ankles?

Visit a healthcare provider as soon as you notice changes or pain in your ankles. Some of the most common tests providers use to diagnose issues include:

X-rays.

Ultrasounds.

Joint aspiration.

CT scans.

MRIs.

Blood tests.

Care

What can I do to relieve ankle pain?

Don't play sports or do any intense physical activity that can put more stress on your ankle until a healthcare provider says it's safe. Follow the RICE method as soon as you notice pain or other symptoms:

Rest: Avoid the activity that caused your injury. Don't overuse your ankle while it heals.

Ice: Apply a cold compress or ice packs wrapped in a thin towel to your ankle for 15 minutes at a time, a few times a day.

Compression: You can wrap your ankle in an elastic bandage to help reduce swelling.

Elevation: Prop your ankle above the level of your heart as often as you can.

Visit a healthcare provider if the pain is severe or your ankle still hurts after a few days of at-home care.